

# Maximising Edge Density in Graphs Subject to Connectivity Constraints

**Erdős Problem 915, from Proof to Search**

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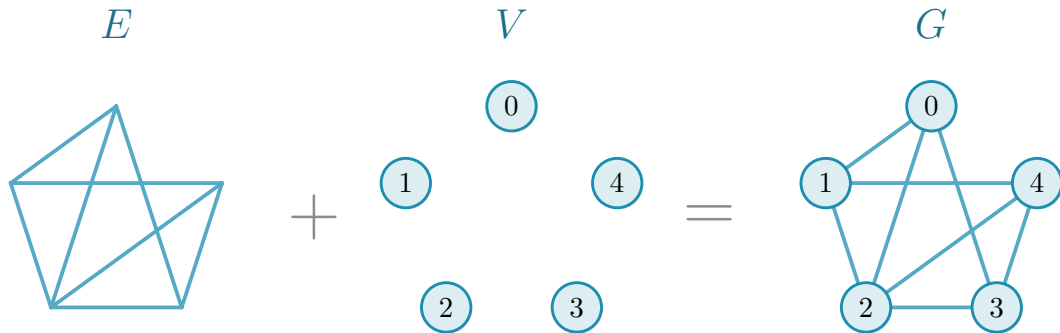
Maximising edge density in graphs  
subject to connectivity constraints



*translation*

networks with **many links** and **low connectivity**

# Graph

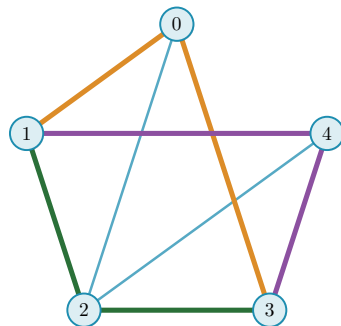


# Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$  = the largest number of  $u$ - $v$  routes that share **no edge**: one colour each

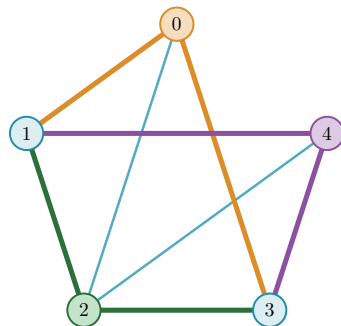


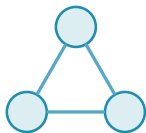
# The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \kappa_G^{\max} \leq m-1 \right\}$$

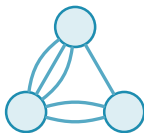
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$  = the largest number of  $u$ - $v$  routes that share **no interior vertex**

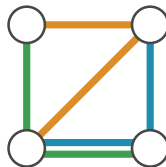




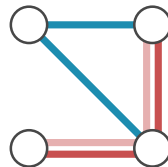
*simple graph*



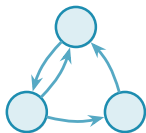
*multigraph*



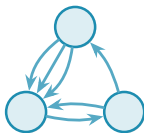
*hypergraph*



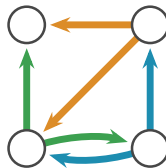
*multihypergraph*



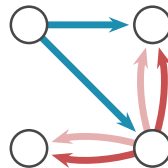
*simple digraph*



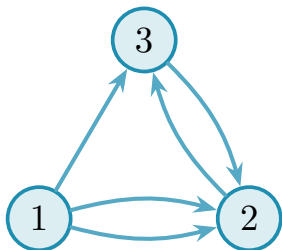
*directed multigraph*



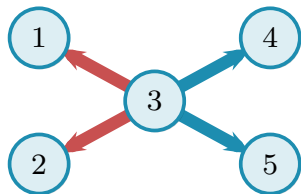
*directed hypergraph*



*directed multihypergraph*



$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

*directed multigraph*

**max flow**  
over every pair

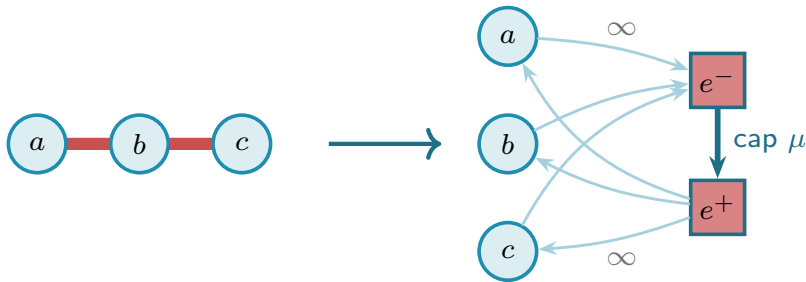
$$\lambda_G^{\max} = 3$$

OUTPUT

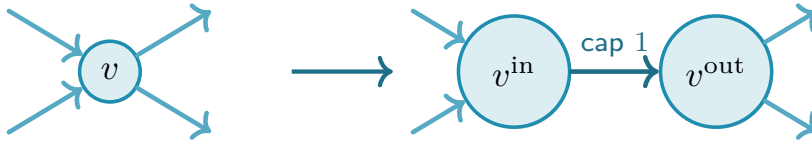
# undirected $\rightarrow$ directed



# hypergraph $\rightarrow$ non-hypergraph



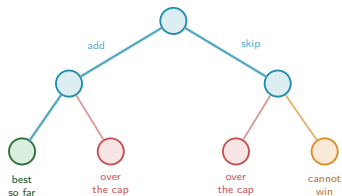
$$\kappa \rightarrow \lambda$$



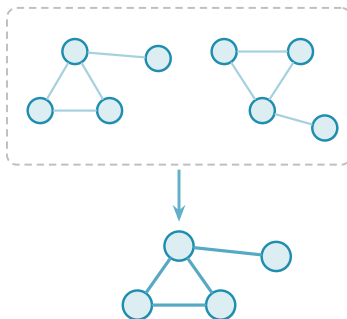
every intermediate vertex is split this way  
the flow itself runs from  $s^{\text{out}}$  to  $t^{\text{in}}$

# Efficiency: three ways to make the sweep finish

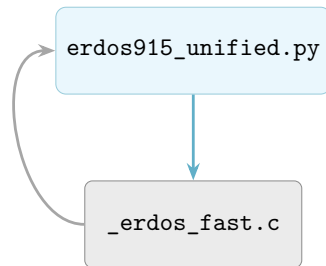
## pruning



## generating search

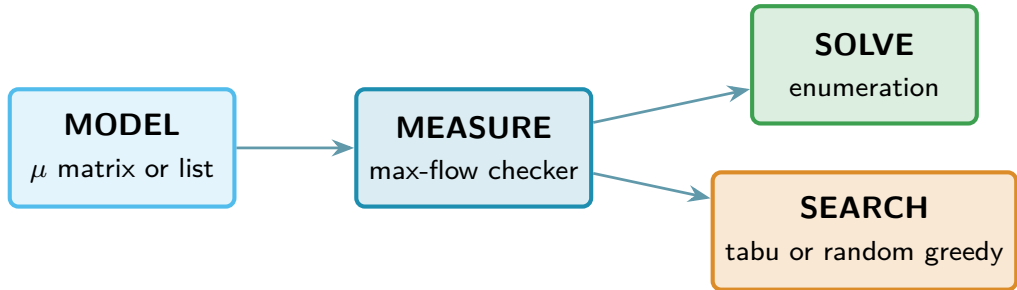


## C: faster max flow



# Solving versus searching

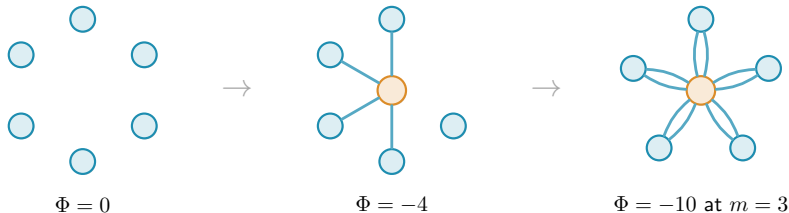
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# The search: one number per graph

$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$

$c^{\max}$  is  $\lambda^{\max}$  or  $\kappa^{\max}$ , according to the separation, and  $\Lambda > 0$  is the penalty weight.

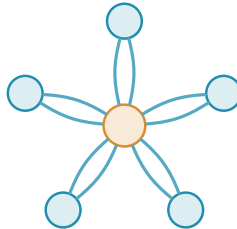


# Extremal families the search rediscovers



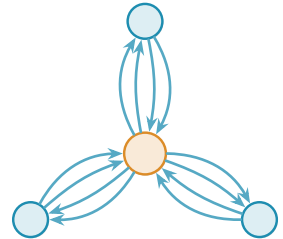
*tree*

$$\ell_2(6) = 5$$



*multigraph star*

$$L_3(6) = 10$$

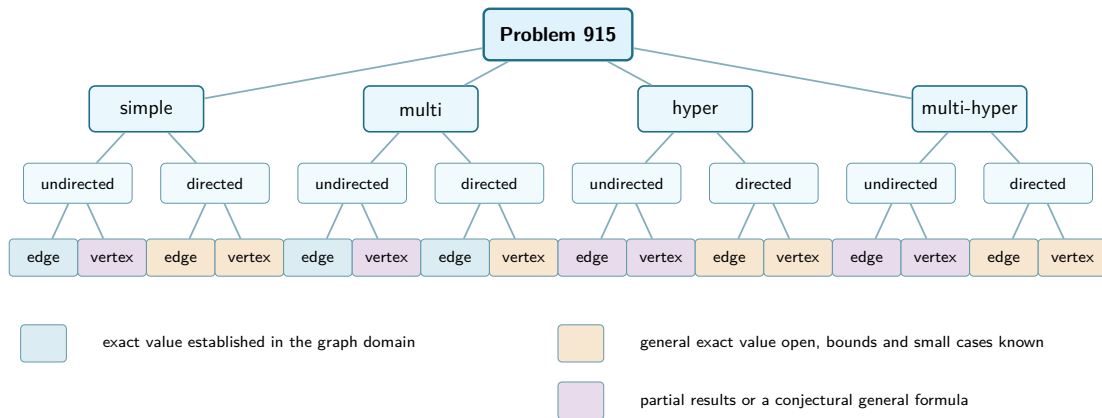


*bidirected star, doubled arcs*

$$L_3^{\text{dir}}(4) = 12$$

examples of the families, not the saved search witnesses

# Status of the sixteen variants



# What this thesis proves

## 1 variant settled completely

exact for every  $n$  and  $m$ , first proved here

directed multigraph · arc

$$L_m^{\text{dir}}(n) = (m-1) \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$$

## 3 variants not new here

settled or studied before

simple graph · edge Mader

simple graph · vertex classical

multigraph · edge  $(m-1)(n-1)$

## 12 variants settled partially

first results here, general exact value still open

multigraph · vertex

$m=2, 3$  exact

(multi)hypergraph · edge (2)

bound, attainment

(multi)hypergraph · vertex (2)

$m=2$  exact,  $m=3$  bound

simple digraph (2)

$m=2$  exact,  $\frac{n^2}{4} + \Theta_m(n)$

directed multigraph · vertex

$m=2$  exact, leading term

directed (multi)hypergraph (4)

leading term

# AI: from reviewer to worker

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**Scale of output = double-edged sword**

**Numerical reliability**

**Parrot proofs**

**AI = deceiving tool**

**Master's thesis = Understanding frontier research**

**Applied + pure mathematics**

**$\forall m, n \exists 32$  problems**

# Thank you

`github.com/louisvandenbruwaene/thesis`

# Backup: the sixteen variants

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## HOW TO READ THE NEXT SIXTEEN FRAMES

Each frame gives bounds and a small witness for one variant. The badge at top right matches the status tree.

**EXACT VALUE ESTABLISHED**

**PARTIAL OR CONJECTURAL**

**GENERAL VALUE OPEN**

**UPPER** the formula, with any condition it needs

**LOWER** what the construction achieves, as a formula too

**PROVED** a checked or cited mathematical bound

**PROPOSED** a conjecture whose proposed proof the author has not checked line by line

**COMPUTED** a finite optimum established by completed enumeration

**1. 2. 3.** a sketch of the bounds and construction

**EXTREMAL** a witness attaining the optimum at the displayed parameters

# Simple graph · edge

$$\ell_m(n)$$

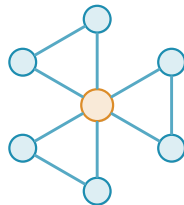
UPPER  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  **PROVED**

LOWER  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  **PROVED**  $n \geq m \geq 2$

1. a Gomory–Hu tree holds every pairwise connectivity in  $n - 1$  cuts
2. their total capacity is at most  $(m - 1)(n - 1)$
3. at most  $n - 1$  edges cross only one tree cut, and all others cross at least two, so  $2|E| - (n - 1) \leq (m - 1)(n - 1)$
4. a hub with an almost regular spoke graph of maximum degree  $m - 2$  attains the bound when  $n \geq m$

**USES** Mader 1.2 · Gomory–Hu A.2 · double count A.5–7 · hub-and-spokes A.8–10

EXACT VALUE ESTABLISHED



$$n = 7, m = 3 \quad 9$$

**EXTREMAL**

# Simple graph · vertex

$$k_m(n)$$

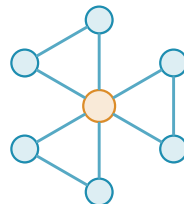
UPPER  $k_m(n) = \ell_m(n)$  **PROVED**  $m \leq 4$

LOWER  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  **PROVED**  $n \geq m \geq 2$

1. the edge bound does not transfer:  $\kappa \leq \lambda$  makes vertex feasibility the weaker restriction, so there are more graphs to beat
2. for  $m \leq 4$  the two values are equal anyway, which is the cited result
3. at  $m = 5$  they part:  $\left\lfloor \frac{8n}{3} \right\rfloor - 4$  takes over from  $n \geq 14$
4. the same drawing attains it, every spoke vertex having degree two

**USES** Bártfai–Bollobás–Leonard 1.4 · Sørensen–Thomassen 1.5 · degree bound A.3

PARTIAL OR CONJECTURAL



$n = 7, m = 3$  **9** **EXTREMAL**

# Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

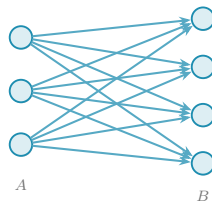
**UPPER**  $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$  **PROVED**  $m = 2$

**LOWER**  $M(n)$  **PROVED**

1. a reachability skeleton is a subdigraph preserving who reaches whom
2. at  $m = 2$  a feasible digraph is its own skeleton: drop an arc and the route replacing it is a second disjoint one, breaching the cap
3. a skeleton carries at most  $M(n)$  arcs, which bounds the digraph
4. at  $n = 7$  both branches of  $M$  equal 12, so the bidirected hub and the one-way bipartite digraph each attain it

**USES** Arc value at  $m = 2$  1.6 · skeleton A.18–A.19 · the two  $m = 2$  families A.22 · directed hub A.15 · one-way bipartite A.16

GENERAL VALUE OPEN



$n = 7, m = 2$  **12**

**EXTREMAL**

# Simple digraph · vertex

$$k_m^{\text{dir}}(n)$$

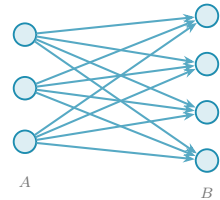
UPPER  $M(n)$  **PROVED**  $m = 2$

LOWER  $M(n)$  **PROVED**

1. the routes the skeleton argument produces are internally vertex-disjoint, not merely arc-disjoint
2. so the argument proves the *vertex* bound first, and the arc bound is the consequence rather than the source
3. that direction matters: Whitney would not license carrying a bound the other way
4. the same digraph attains both, so the two values coincide at  $m = 2$

**USES** Vertex value at  $m = 2$  1.7 · sparse skeleton A.19 · deletion monotonicity A.20 · one-way bipartite A.16

GENERAL VALUE OPEN



$n = 7, m = 2$  **12**

**EXTREMAL**

# Multigraph · edge

$$L_m(n)$$

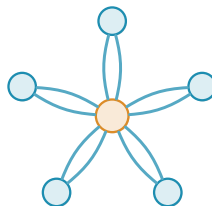
**UPPER**  $(m-1)(n-1)$  **PROVED** every  $n$  and  $m$

**LOWER**  $(m-1)(n-1)$  **PROVED**

1. the same Gomory–Hu charge:  $n-1$  cuts, each of capacity at most  $m-1$
2. each edge copy crosses at least one cut, so  $|E| \leq (m-1)(n-1)$
3. parallel copies invalidate the simple graph's bound of  $n-1$  distance-one edges, so that stronger double count is unavailable
4. any spanning tree at multiplicity  $m-1$  attains it

**USES** Multigraph edge value and thickened-tree proof 1.3 · Gomory–Hu A.2

**EXACT VALUE ESTABLISHED**



$$n = 6, m = 3 \quad \mathbf{10}$$

**EXTREMAL**

# Multigraph · vertex

$$K_m(n)$$

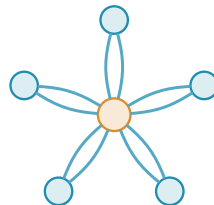
UPPER  $2(n-1)$  **PROVED**  $m=3$

LOWER  $(m-1)(n-1)$  **PROVED**

1. between two vertices  $\kappa$  is exactly  $\mu(u, v) + \pi(u, v)$ : the copies of their own edge, plus the disjoint detours around it
2. feasibility therefore caps those two *together* at  $m-1$
3. at  $m=3$  the incidence-rank lemma gives the same value as the edge problem
4. a tree edge has no detour, so a doubled tree spends the whole cap on multiplicity and attains it

**USES** Splitting the vertex measure A.51 · value at  $m=2$  A.52 · value at  $m=3$  A.54 · incidence rank A.49

PARTIAL OR CONJECTURAL



$n=6, m=3$  **10**

**EXTREMAL**

# Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

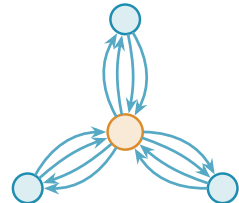
**UPPER**  $(m - 1)M(n)$  **PROVED** every  $n$  and  $m$

**LOWER**  $(m - 1)M(n)$  **PROVED**

1. inside any feasible multigraph there is a sparse reachability skeleton
2. removing one copy of each of its arcs costs every reachable pair at least one route, so the graph stays feasible one threshold lower
3.  $m - 1$  such peels account for every arc, giving  $m - 1$  times the  $m = 2$  value, and the integrality of the fractional relaxation follows
4. at multiplicity  $m - 1$ , a bidirected tree attains the linear branch and a one-way bipartite digraph the quadratic one

**USES** Every  $n$  and  $m$  1.13 · sparse skeleton A.19 · deleting a skeleton A.34 · thickened tree and bipartite digraph A.33 · integrality A.36

**EXACT VALUE ESTABLISHED**



$n = 4, m = 3$  **12**

**EXTREMAL**

# Directed multigraph · vertex

$$K_m^{\text{dir}}(n)$$

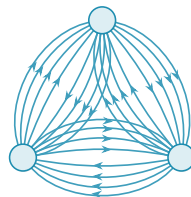
**UPPER**  $(m-1) \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1)$  **PROVED** reads 210  
here, far from tight

**LOWER**  $n(n-1)(m+1-n)$  **PROPOSED**  $2 \leq n \leq m$

1. the proposed general argument splits  $\kappa$  into direct copies plus internally disjoint detours
2. where all six directions are present each pair has exactly one detour, so its multiplicity stops at  $m-2=4$  rather than at the cap of five
3. the six directions give  $6 \times 4 = 24$ , and enumeration leaves exactly one isomorphism class
4. it is A.62's thickened complete digraph at multiplicity 4

**USES** Directed incidence value A.61 · proposed splitting A.60 · construction A.62 and proposed bound A.63

GENERAL VALUE OPEN



$$n = 3, m = 6 \quad \mathbf{24}$$

COMPUTED

EXTREMAL

# Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

UPPER  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED**

LOWER  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED**  $m-1 \leq \binom{n-2}{r-2}$

1. split along a cut of at most  $m-1$  hyperedge copies and induct on the two trace hypergraphs
2. each crossing copy adds one to the trace sums of  $|e|-1$ , hence  $(r-1)|E| \leq (m-1)(n-1)$
3. the reduction to graphs points the wrong way: replacing a hyperedge by a star would let two routes take two of its edges, which Berge forbids
4. the star hypertree attains the illustrated  $m=2$  case, and a hub with degree-capped hyperedges attains the stated general range

**USES** Hypergraph edge bound 1.8 · cut induction A.42 · simplicity is free A.43 · sparse hypergraphs A.46

PARTIAL OR CONJECTURAL



$$n=9, r=3, m=2 \quad 4$$

EXTREMAL

# Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

UPPER  $\left\lfloor \frac{n-1}{r-1} \right\rfloor$  **PROVED**  $m = 2$

LOWER  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED**  $r \geq 3, m-1 \leq \binom{n-2}{r-2}$

1. at  $m = 2$  feasibility says exactly that the incidence graph is a forest, since a cycle in it would already be two disjoint routes
2. a forest on the  $n$  vertices together with the  $|E|$  hyperedges has at most  $n + |E| - 1$  edges
3. each hyperedge contributes  $r$  of them, so  $r|E| \leq n + |E| - 1$ , which rearranges to  $|E|(r-1) \leq n-1$
4. the star hypertree's incidence graph is a tree, so it attains it

**USES** Vertex value at  $m = 2$  1.9 · at  $m = 3$  1.10 · general lower bound A.47 · incidence rank A.49 · Menger for multihypergraphs A.41

PARTIAL OR CONJECTURAL



$$n = 9, r = 3, m = 2 \quad 4$$

**EXTREMAL**

# Multihypergraph · edge

$$L_m^{(r)}(n)$$

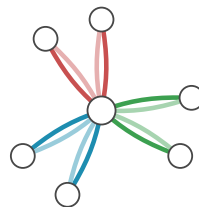
**UPPER**  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED** unchanged by repetition

**LOWER**  $\frac{(m-1)(n-1)}{r-1}$  **PROVED**  $(r-1) \mid (n-1)$

1. the cut induction never asks whether hyperedges are distinct
2. so the bound is word for word the one next door, and repetition cannot move it
3. what repetition changes is which shapes reach it: the divisibility condition replaces the supply of distinct hyperedges
4. the star hypertree with each hyperedge taken  $m-1$  times attains it

**USES** Hypergraph edge bound 1.8 · cut induction A.42 · multihypergraph lower bound A.48 · twelve copies on six vertices A.45

PARTIAL OR CONJECTURAL



$n = 7, r = 3, m = 3$  **6**

**EXTREMAL**

# Multihypergraph · vertex

$$K_m^{(r)}(n)$$

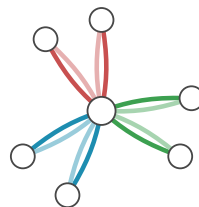
UPPER  $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$  **PROVED**  $m = 3$

LOWER  $\frac{(m-1)(n-1)}{r-1}$  **PROVED**  $(r-1) \mid (n-1)$

1. the incidence graph again, but now its cycle rank is what needs bounding
2. the rank is controlled by decomposing the graph at its cut vertices and at its separating pairs
3. repetition buys nothing at  $m = 2$ , where two copies of one hyperedge are already two separated routes
4. from  $m = 3$  it adds attainment cases, and the repeated star is one

**USES** Vertex value at  $m = 3$  1.10 · incidence rank A.49 · multihypergraph lower bound A.48

PARTIAL OR CONJECTURAL



$$n = 7, r = 3, m = 3 \quad \mathbf{6}$$

**EXTREMAL**

# Directed simple hypergraph · arc

$$\ell_m^{\text{dir}(r)}(n)$$

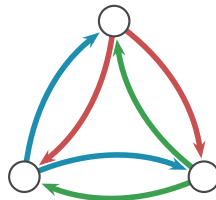
**UPPER**  $\frac{(m-1)n(n-1)}{r-1}$  **PROVED** reads 6 here, not tight

**LOWER**  $n \cdot \min(m-1, \binom{n-1}{r-1})$  **PROVED**  $n \geq r$

1. every route out of a vertex has to start with one of that vertex's own outgoing hyperedges
2. deleting those cuts it off, so the outgoing count alone caps every pair it reaches, and giving each vertex  $\min(m-1, \binom{n-1}{r-1})$  of them is feasible at any size
3. for large  $n$  a separate bipartite construction is asymptotically optimal; a shadow digraph supplies its upper bound
4. at this size enumeration closes the gap, and  $3 \times 1 = 3$

**USES** Directed hypergraph constant 1.14 · first bounds A.64 · bounded outgoing count A.65 · one-step shadow A.68 · two-step bound A.29

GENERAL VALUE OPEN



$n = 3, r = 3, m = 3$  **3**

COMPUTED

EXTREMAL

# Directed simple hypergraph · vertex

$$k_m^{\text{dir}(r)}(n)$$

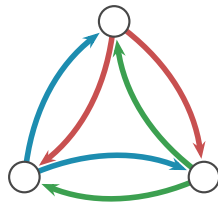
**UPPER**  $\frac{(m-1)n(n-1)}{r-1}$  **PROVED** reads 6 here, not tight

**LOWER**  $n \cdot \min(m-1, \binom{n-1}{r-1})$  **PROVED**  $n \geq r$

1. the outgoing cut never looks at interior vertices at all
2. so the very same argument establishes the arc bound and the vertex bound in one go, and the witness needs no adjustment
3. the shadow argument behind the general growth rate re-runs under the stricter separation for the same reason
4. whether the two separations keep one value at larger sizes is open

**USES** Directed hypergraph constant 1.14 · first bounds A.64 · bounded outgoing count A.65

GENERAL VALUE OPEN



$n = 3, r = 3, m = 3$  **3**

**COMPUTED**

**EXTREMAL**

# Directed multihypergraph · arc

$$L_m^{\text{dir}(r)}(n)$$

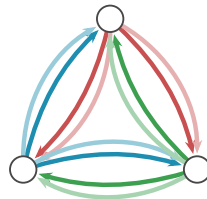
UPPER  $\frac{(m-1)n(n-1)}{r-1}$  **PROVED**

LOWER  $n(m-1)$  **PROVED**  $n \geq r$

1. counting tail-head pairs gives the upper bound: each hyperedge covers  $r-1$  of them, and no ordered pair takes more than  $m-1$
2. give each tail  $m-1$  copies: every route from it uses a distinct outgoing copy, so this construction is feasible and reaches  $n(m-1)$
3. at this size a tail has only one possible head set, the other two vertices, so repetition is the only way to grow at all
4. the two meet here, both reading 6, so the value needs no search:  $n(m-1)$  equals the counting bound exactly when  $n = r$

**USES** Directed hypergraph constant 1.14 · first bounds A.64 · bounded outgoing count A.65 · backward reverses forward A.66

GENERAL VALUE OPEN



$$n = 3, r = 3, m = 3 \quad \mathbf{6}$$

EXTREMAL

# Directed multihypergraph · vertex

$$K_m^{\text{dir}(r)}(n)$$

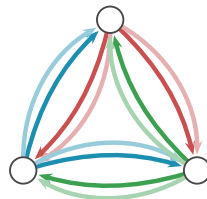
UPPER  $\frac{(m-1)n(n-1)}{r-1}$  **PROVED**

LOWER  $n(m-1)$  **PROVED**  $n \geq r$

1. the pair count bounds this variant too, and the outgoing cut ignores interior vertices, so both bounds carry over from the arc problem unchanged
2. they meet here as well, so this size is settled by argument rather than by search
3. the general model, which allows every non-empty tail–head split, obeys the forward bound and shares its constant
4. whether forward and general agree at a fixed  $n > r$ , and not merely in the growth rate, is open

**USES** General orientation constant A.69 · general obeys forward A.67 · backward reverses forward A.66 · bounded outgoing count A.65

GENERAL VALUE OPEN



$$n = 3, r = 3, m = 3 \quad \mathbf{6}$$

**EXTREMAL**

# An extremal directed multihypergraph

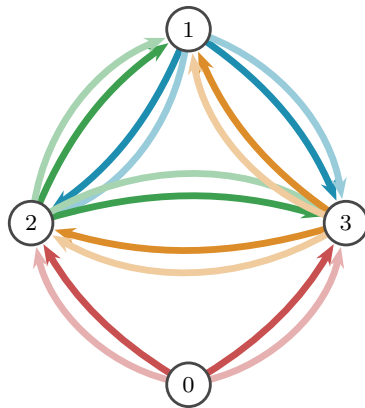
$n = 4$ ,  $r = 3$ ,  $m = 3$ , arc separation

$$L_3^{\text{dir}(3)}(4) = 8$$

COMPUTED

EXTREMAL

- exhaustive search over all  $3^{12} = 531\,441$  ways to give the 12 possible hyperedges a multiplicity in  $\{0, 1, 2\}$ , in 593 s
- every vertex is the tail of one hyperedge, taken twice: the bounded outgoing count  $n(m-1) = 8$  of A.65 is optimal here
- $\lambda^{\max} = 2$ : every route out of a vertex starts in one of its own two copies
- the counting bound  $\frac{(m-1)n(n-1)}{r-1} = 12$  is not attained



one colour per hyperedge, light shade =  
second copy